Judge-Guided Generation, Selection and Repair of Inline Non-Verbal Sounds in Vietnamese Speech Synthesis

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Abstract

We describe the WiTake system for the VLSP 2026 ViNV-TTS shared task: Vietnamese text-to-speech in which inline tags (`[laughter]`, `[breathing]`, `[sniff]`, `[throatclearing]`) must be realised as the corresponding sound at the tagged position, both for speakers seen in training (Track A) and for zero-shot cloning from a reference clip (Track B). The system starts from the open masked non-autoregressive TTS model OmniVoice and adds (i) supervised fine-tuning on the official data plus synthetic rare-class events, (ii) a class-aware preference optimisation (DPO) stage adapted to masked generative TTS, (iii) *tag-contrastive* classifier-free guidance that contrasts the tagged text with the same text without tags, (iv) best-of-16 selection over two engines with automatic judges weighted like our proxy of the official score, (v) splicing of real non-verbal clips where the judges still find a tag missing, and (vi) a loudness-levelling step that maps generated non-verbal events onto the level distribution of real recordings. On the public test set, tag-contrastive guidance alone raises the hit rate of a held-out tag-ASR judge from .58 to .83 (Track A, single draw). The full system reaches .93/.96 (Track A/B) on that judge, which is optimistic because the judge also drives selection, and about .89 on an independent audit judge, with automatic WER of 1.5/1.7% and unchanged naturalness proxies. The system is packaged as an offline single-GPU Docker image. We also report the many things that did not help.

1 Introduction

Conversational speech is full of non-verbal vocalisations (NVs): laughter, audible breaths, sniffs, throat clearing. Modern zero-shot TTS models reproduce the timbre and prosody of a prompt well but rarely produce such events on demand, and when an NV tag is given in the text the model often ignores it or reads it as noise. The ViNV-TTS task asks for Vietnamese speech in which four NV classes are produced exactly where an inline tag places them.

Three properties of the task shaped our system. First, the classes are very unbalanced: the official training set has 3,733 breathing tags but only 117 sniff and 115 throat-clearing tags (Table 1). Second, NV placement accuracy is the largest single term of the official score (weight .3, against .1–.2 for every other term), so a missed tag costs more than a small loss in naturalness. Third, NV realisation is cheap to *check* automatically (an aligned word gap can be classified) but hard to *guarantee* in one sampling pass. We therefore treat generation as a generate–judge–repair loop: make the model more likely to produce the event (fine-tuning, DPO, guidance), sample several candidates and pick the best with judges, and repair what is still missing.

Our contributions are: a tag-contrastive guidance term for masked generative TTS that raised the held-out tag hit rate from .58 to .83 at no training cost; a class-aware DPO recipe for masked non-autoregressive TTS; a selection-and-splice pipeline whose gains we verify with judges that were never used for selection; an NV loudness-levelling step that corrects the over-loud events that every detectability-driven component produces; and an honest list of negative results.

2 Task and Data

Task. Each test item gives a target text with inline tags and a reference (prompt) clip with its transcript. Track A uses speakers from the training set, Track B unseen speakers. The official score[1] combines NV placement accuracy (NVPA), human-rated naturalness (SN) and quality (Q), a human SUS-test WER, an automatic WER (aWER, Zipformer) and speaker similarity (SS, ECAPA-

[1] https://vlsp.org.vn/vlsp2026/eval/vinvtts

split	utts	spk	hours	breath	laugh	sniff / throat
train	1739	189	5.90	3733	681	117 / 115
dev	316	46	1.04	674	102	14 / 19

Table 1: Official ViNV-TTS data (tag counts per class).

TDNN): Track A $= .3\,\mathrm{NVPA} + .2\,\mathrm{SN} + .2\,\mathrm{Q} + .1(1{-}\mathrm{WER}){+}.1(1{-}\mathrm{aWER}){+}.1\,\mathrm{SS}$, and Track B $= .3\,\mathrm{NVPA} + .15\,\mathrm{SN} + .15\,\mathrm{Q} + .1(1{-}\mathrm{WER}) + .1(1{-}\mathrm{aWER}) + .2\,\mathrm{SS}$.

Automatic proxy. During development we optimised an automatic proxy with four terms and the weights we used throughout development (Track A: NV .3, WER .15, MOS .15, SS .1; Track B: NV .3, WER .1, MOS .1, SS .2), normalised by their sum:

$$\mathit{off}_T = \tfrac{1}{0.7}\big(.3\,\mathrm{NV} + w^T_{\mathrm{wer}}(1{-}\mathrm{WER}) + w^T_{\mathrm{mos}}\tfrac{\mathrm{pMOS}}{5} + w^T_{\mathrm{ss}}\,\mathrm{SS}\big)$$

with WER from a Vietnamese Zipformer transducer (Yao et al., 2024; hynt and csukuangfj, 2026), pMOS = DNSMOS OVRL (Reddy et al., 2022), SS = ECAPA-TDNN cosine similarity to the prompt (Desplanques et al., 2020; Ravanelli et al., 2021), and NV = hit rate of our tag-ASR judge v3 (Section 3.6). We write *offA*/*offB* for Track A/B. Human-rated terms are not included.

Data. We used the official training and development sets (Table 1), VocalSound (Gong et al., 2022) as a source of sniff and throat-clearing events, and, as text only, sentences transcribed from public Vietnamese football talk videos for preference-optimisation prompts. The public test set has 210 Track A items (7 speakers) and 120 Track B items (4 unseen speakers); with 32–56 tags per class, differences of about .03 in a per-class hit rate are within noise. For model selection we also held out 114 training utterances (*evalR*).

3 System

Figure 1 shows the pipeline. Two fine-tuned OmniVoice engines generate 8 candidates each; every candidate is scored by automatic judges; the best one under the proxy-weighted utility is kept; missing events are spliced in from real recordings; and NV loudness is levelled.

3.1 Base model

OmniVoice (k2-fsa, 2026) is a masked generative (non-autoregressive) TTS model, like MaskGCT

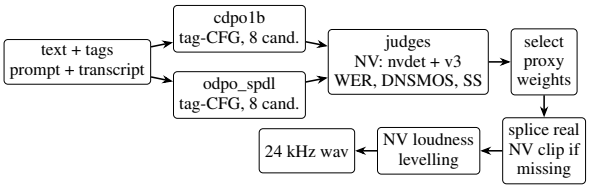


Figure 1: Inference pipeline (v5). In our public-test runs, Track A additionally re-generates ("inpaints") missed tags inside the primary engine before selection; the submitted image runs every item with the Track B settings.

(Wang et al., 2024): a transformer predicts the masked audio tokens of a multi-codebook codec (the Higgs Audio 2 tokenizer) conditioned on text, the prompt transcript and the prompt's audio tokens, and fills the sequence in 32 iterative steps with classifier-free guidance. Its zero-shot cloning was clearly stronger than the alternatives we tried (Section 5). Out of the box it ignores NV tags.

3.2 Supervised fine-tuning with synthetic rare events

We fine-tune OmniVoice with LoRA (Hu et al., 2022) using the official recipe on the training set, with the tags kept as text and rare classes oversampled (square-root sampling). Because sniff and throat clearing have only about 115 examples each, we generate synthetic examples: VocalSound events are converted to the voice of a training speaker with Seed-VC (Plachtaa, 2024) and inserted into that speaker's utterances at a word gap, then filtered with our rare-class judge. We also add laughter clips generated by an earlier fine-tuned TTS model and filtered with the laughter judge and WER $\leq .10$. The resulting model (*omft_tdl4000*, 4,000 steps) uses the synthetic throat-clearing clips but no synthetic sniffs; it is the parent of both served engines. Its original synthetic throat clips were not kept, so the released set (277 clips: 174 throat clearing, 103 sniff) is a regenerated one. Continuing fine-tuning from omft_tdl4000 on the official data alone, without the synthetic and laughter clips, cut throat recall on evalR from .69 to .28.

3.3 Class-aware DPO for masked generative TTS

Fine-tuning made the model capable of the events but did not make it reliable on rare classes. We therefore applied on-policy preference optimisation (Rafailov et al., 2023). Masked generative

models have no tractable sequence likelihood, so we use the masked cross-entropy as a likelihood surrogate: for a target token sequence y, $\ell(y)$ is the codebook-weighted mean cross-entropy on masked positions, averaged over $K{=}4$ stratified mask ratios, with the policy and the frozen reference sharing every mask. The loss is

$$\mathcal{L} = -\log\sigma\big(\beta[(\ell_{\text{ref}}(y_w){-}\ell_\theta(y_w)) - (\ell_{\text{ref}}(y_l){-}\ell_\theta(y_l))]\big) + \gamma\,\ell_\theta(y_w)$$

where the last term anchors the policy to the chosen sample.

Pairs. We built 1,921 prompts (1,660 football-talk sentences with one or two inserted tags, 15% of them sentence-initial and 15% sentence-final; 261 single-class training texts), each with a 3–9 s reference clip of a training speaker. No public-test text or audio was used. The SFT model generated 8 candidates per prompt (15,368 in total). Each tag was judged by the AST-based detector and a class-specific judge (Section 3.6). The chosen sample realises every tag with WER $\leq .15$ and SS above the prompt median minus .05; the rejected sample is a candidate of the same prompt in which the target-class tag is missing. We kept at most one pair per prompt and balanced the classes (649 pairs: breathing 179, throat 179, laughter 153, sniff 138), then trained for 1,298 single-pair steps (twice the pair count, pairs sampled with replacement, gradient accumulation 8, $\beta{=}5$, lr $5 \cdot 10^{-5}$, LoRA rank 16) and merged the adapter (*cdpo1b*). The tag-ASR judge v3 was never used to build pairs, so it remains a held-out judge for this step.

The second engine, *odpo_spdl*, comes from a separate SFT run that also uses the synthetic sniff clips followed by DPO on 600 training prompts $\times$ 4 on-policy candidates, ranked by an earlier Whisper-based tag-ASR (Radford et al., 2023) and the AST detector.

3.4 Tag-contrastive guidance (tag-CFG)

Standard classifier-free guidance (Ho and Salimans, 2022) contrasts the conditional logits c with unconditional ones u. We add a second contrast against the logits $c_{\setminus\text{tag}}$ obtained with the same text *with the tags removed*:

$$\log p = c + g\,(c - u) + w\,(c - c_{\setminus\text{tag}}).$$

The extra term amplifies exactly what the tags change in the prediction, i.e. the NV events. All three rows are computed in one batched forward pass per decoding step. We use $g{=}2, w{=}3$ for Track A and $g{=}6, w{=}5$ for Track B on the primary engine, chosen on a partial grid over $g \in \{2, 4, 6\}$ and $w \in \{0, 3, 5, 7\}$ run on the SFT parent.

3.5 Verify-and-inpaint (Track A)

For Track A the primary engine checks each candidate after decoding: Zipformer word timestamps locate the gap of every tag, the AST detector classifies it, and for every missed tag the audio tokens of that gap ($\pm$0.12 s plus a few inserted frames) are re-masked and regenerated with all other tokens fixed, for up to $k{=}2$ rounds. Inpainting all 8 candidates gives the selector more tag-complete candidates. Repairing only the selected candidate kept almost none of the gain (+.002 vs +.020 NV measured by nvdet, which also decides what to inpaint).

3.6 Judges and selection

- **nvdet**: AST (Gong et al., 2021) embeddings of the aligned tag gap with a logistic-regression head over {none, 4 classes}, trained on events cut from the official training set.
- **tag-ASR v3**: Qwen3-ASR-1.7B (Qwen Team, 2026) fine-tuned to transcribe the training set *with* inline tags. A tag counts as realised when the transcript contains it at the right position. Dev precision/recall: breathing .90/.89, laughter .72/.82, but sniff .36/.29 and throat .60/.32.
- **Class judges**: LaughterSegmentation (Omine et al., 2024) for laughter, and frame-level PretrainedSED models (Schmid et al., 2024) (BEATs, ATST-F, M2D) with small logistic heads for sniff and throat clearing.
- **Audit judge** (evaluation only): CLAP (Wu et al., 2023) + WavLM (Chen et al., 2022) features with a logistic head. It shares no model with the other judges and was never used by generation, selection or splicing.

Each of the 16 candidates gets the proxy-weighted utility off_T with NV = mean over tags of $\frac{1}{2}$(nvdet p_{tag} + v3 hit). In a leave-one-judge-out study, nvdet alone as the selector overfitted as N grew (held-out v3 NV on Track B .83$\rightarrow$.79 from $N{=}8$ to 24), while the nvdet+v3 average was the most stable rule (held-out class-judge NV .91–.98).

3.7 Splicing real NV clips

If, on the selected candidate, the tag-ASR or the class judge still finds a tag missing (decision rule *any*), the tag's word gap is replaced by 80 ms pause + a real NV clip + 120 ms pause with 15 ms cross-fades. Clips come from a bank of events cut from the official training set (breathing 1,021, laughter 337, sniff 65, throat 80), filtered by nvdet confidence, SNR and length. For Track A the clip comes from the same speaker when available; for Track B from the three ECAPA-nearest bank speakers. Converting bank clips to the prompt voice with Seed-VC removed the NV character (nvdet hit rate .83→.31) for +.002 SS, so we splice unconverted clips.

3.8 NV loudness levelling

Tag-CFG, judge-ranked DPO, best-of-N and splicing all reward *detectable* events. Measured relative to the surrounding speech, our laughter and throat clearing sat about 3–4 dB and our sniffs 9–12 dB louder than in real recordings. For each class we map the system's NV-to-speech level distribution onto the distribution measured on the training set with a 1-D quantile (optimal-transport) map, and only ever attenuate the aligned NV span, with raised-cosine ramps (throat clearing is moved only 60% of the way, α=.6; breathing is untouched). On the submitted outputs this reduced the Wasserstein-1 distance between system and real NV levels from 5.9 to 3.6 dB on Track A and from 9.0 to 5.6 dB on Track B, with WER, pMOS and SS unchanged.

3.9 Deployment

The organisers ran each image offline on one 40 GB GPU with a 60-minute limit for about 50 items. Our image contains all weights and four Python environments; per-process GPU memory caps keep the five services below 35 GB (peak 34 GB), and the per-candidate count steps down (8→4→2→1 per engine) if the measured speed projects past the time budget. Every generation call is seeded from its text, so reruns are equivalent (GPU kernels are not bit-exact). The image runs every item with the Track B settings (reference-prompt cloning, Track B guidance and weights, ECAPA-nearest splice clips); verify-and-inpaint and same-speaker clips were used only in our public-test runs. On one H100 a 50-item test takes 7.6 min including 20 s of start-up.

config	NV	sniff	throat	WER	pMOS	SS
plain (w=0)	.580	.32	.59	.027	3.28	.731
w=1	.759	.52	.77	.026	3.27	.732
w=3	.826	.59	.80	.025	3.24	.729
w=5	.897	.71	.95	.029	3.21	.734
inpaint k=3 only	.750	.50	.86	.027	3.28	.731
w=3 + inpaint k=2	.884	.66	.96	.026	3.25	.732

Table 2: Tag-CFG and inpainting on public Track A (SFT model, single draw, NV = tag-ASR v3 macro hit rate).

decide	A: NV	A: offA	B: NV	B: offB
no splice	.594	.708	.633	.724
nvdet	.754	.775	.750	.775
v3	.817	.801	.828	.813
any (v3 or class judge)	.799	.793	.828	.813
any + nvdet	.781	.786	.781	.794

Table 3: NV splicing on plain SFT outputs (public test). NV = tag-ASR v3 hit rate; in the v3 and *any* rows v3 also decided which tags to splice.

4 Results

Tag-CFG and inpainting. Table 2 isolates the guidance term on the SFT model (single draw, no selection). Tag-CFG lifts the held-out v3 hit rate from .580 to .826 at w=3 and .897 at w=5, with unchanged WER and SS and a small pMOS cost; inpainting adds a further gain on top.

Splicing. Table 3 shows splicing on the plain SFT outputs. Every decision rule improves every independent judge, so the gain is not an artefact of deciding with the same judge that scores. Deciding with v3 (alone or with the class judges, *any*) gives the highest NV and offA/offB in the table, but those rows are scored by the judge that made the decisions; on nvdet, the judge independent of them, deciding with nvdet scores higher (Track A .772 vs .743 for *any*). Adding nvdet's flags on top of *any* lowers NV and the proxy on both tracks, because replacing an event that was already right is a chance to do worse. In the full pipeline, with tag-CFG, splicing becomes a safety net that fires on 3% of tags (1% in the submitted v5 run).

DPO. Under the full production recipe, cdpo1b improved over its SFT parent on both tracks (offA .878→.882, offB .870→.875), mostly through sniff (Track B .81→.94) and sentence-final tags on Track B (.79→.96). On evalR, sniff rose from .56 to .74 and throat from .84 to .88. These differences are only one or two tags per class, so we treat them as consistent but small.

system	A: NV	offA	B: NV	offB
SFT, best-of-8, plain gen. + splice	.897	.845	.898	.849
+ tag-CFG w=3	.969	.878	.938	.870
v2: + 2nd engine, nvdet+v3 selector	.946	.868	.953	.877
v3: DPO engine (cdpo1b)	.969	.879	.945	.874
v4: + Track A inpaint k=2	.982	.883	.945	.874
v5: + NV levelling (submitted)	.933	.863	.961	.879
WER / pMOS / SS (v5)	.015 / 3.34 / .764		.017 / 3.33 / .812	

Table 4: Served system on the public test (210 A / 120 B items). In every row v3 also drives selection and splicing, so its NV is optimistic; each run is a single draw (NV noise about ±.03).

System evolution. Table 4 tracks the served system on the public test set. The largest single gain comes from tag-CFG; the second engine, DPO and inpainting move the v3-scored numbers by at most .02 in either direction. The audit judge (never used for selection) confirms the overall gain: its macro hit rate excluding sentence-initial tags rose from .82 (plain) to .94–.95 (v2–v4) on both tracks, but between v2 and v4 its differences are within noise. Levelling (v5) attenuates the over-loud events. This costs automatic NV (audit judge .94→ about .89 on both tracks, v3 on Track A .982→.933), because the detectors hear quieter events less often, but it is aimed at the human-rated naturalness that the proxy does not measure.

Official evaluation. Official results on the hidden test set were not available at the time of writing.

5 What Did Not Help

- **Another base model.** A LoRA-tuned VoxCPM2[2] almost never produced mid-sentence NVs (hit rate ≈.03) and was only +.01 SS over OmniVoice.
- **More in-domain data.** We collected 62 h of Vietnamese football and talk-show audio, labelled by our tag-ASR and filtered by detectors. Every dose of it in continued fine-tuning lowered the public scores (NV .594→.41–.46, offA .708→.63–.66); the automatic labels contained no usable sniff or throat clearing, and breathing dominated.
- **Edge-position fine-tuning.** Oversampling clips cut so that events sit at the sentence edge raised sentence-final sniff but collapsed throat clearing (public .52→.20–.25), because the synthetic throat data could not be replayed.
- **Further DPO rounds.** Rounds 2 and 3 (more prompts, pMOS tie-breaks, multi-objective pairs) gained up to +.012 offB but lost Track A or the audit judge; the gains on judges seen during pair building did not transfer.
- **Clean-label fine-tuning.** Removing 1,007 tags that all judges reject raised the audit judge (+.04–.08) but lowered offA/offB by .002–.019.
- **More candidates.** N=24 instead of 16 changed offA by −.005, within noise, at 2.4× the cost; on Track B a larger N let the nvdet-based selector overfit.
- **Post-hoc voice conversion or enhancement.** Seed-VC raised SS by .03–.05 but blurred NV events (v3 NV .68→.53–.59); speech enhancement removed them.
- **Variants of tag-CFG** (per-class weights, step schedules, localised guidance) did not beat uniform guidance on both NV and pMOS.

6 Discussion

The system depends on its judges in two ways: they rank DPO pairs and candidates, and they decide what to repair. With more selection pressure (larger N, more DPO rounds) the system learns to please the judge it is optimised against, so we kept at least one judge out of every decision and checked each change with a fully independent audit judge. The judges themselves are weakest exactly where the task is hardest: sniff and throat clearing are rare, short and easily confused with breathing, and sentence-initial events are underread by our audit judge. Finally, detectability and naturalness pull in opposite directions; levelling reduced the loudness gap to real recordings but does not remove it.

7 Conclusion

Reliable inline NV synthesis came less from a better base model than from a generate–judge–repair loop around a fine-tuned masked TTS model: tag-contrastive guidance makes events likely, class-aware DPO helps the rare classes, judge-based selection and splicing catch the rest, and levelling restores realistic loudness. Code, training recipes and the released models are listed in our reposi-

[2] https://huggingface.co/openbmb/VoxCPM2

tory[3], together with every pretrained model and dataset used and their licences.

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[3] https://github.com/AndrewNgo-ini/vlsp2026-vinvtts-witake